



RV College of Engineering®

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Question Paper ID: 153195

Course Code: AI364TA

USN 1 2 V 2 3 A 1 0 7 2

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post

Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026

Cloud Computing Technology and Architectures

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

16+

Question No	Question	M	CO	BT
1.1	What is meant by elasticity in cloud computing?	02	2	1
1.2	List any four characteristics of Cloud Computing.	02	1	2
1.3	State any two benefits of the Cloud Reference Model.	02	2	2
1.4	What is Microsoft Hyper-V?	02	1	2
1.5	A distributed application needs asynchronous communication between microservices without direct interaction. Explain why Amazon SQS is preferred in this scenario.	02	4	3
1.6	A distributed storage system stores large files by dividing them into replicated chunks managed by a master server. Identify the storage architecture and explain its purpose.	02	4	2
1.7	A company is migrating workloads to multiple public cloud providers while retaining its on-premises Active Directory. Discuss the role of LDAP and Kerberos in enabling secure authentication.	02	3	3
1.8	A hospital stores patient records on its private data center while hosting its appointment portal on Azure. Identify the deployment model and justify your answer.	02	3	2
1.9	A DevOps team is setting up a CI/CD pipeline for a cloud-native application. Identify any four artifacts that should be maintained under version control and explain why they should be versioned.	02	5	3

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1.10	Illustrate the roles of Data Analytics, Machine Learning, Automation, and Visualization in AIops.	02	5	2
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Part B

Question No	Question	M	CO	BT
2a ✓	Discuss the benefits of cloud computing for modern organizations. Support your answer with suitable examples.	06	2	1
2b ✓	Compare a traditional IT environment with a cloud computing environment in terms of infrastructure, virtualization, resource management, scalability, maintenance, security, and cost. Justify why organizations are increasingly adopting cloud computing.	10	4	1
3a	Compare PaaS-I, PaaS-II, and PaaS-III with respect to their description, product type, and vendor examples. Illustrate your answer	08	2	2
3b	Explain the advantages and disadvantages of virtualization. Discuss how virtualization influences resource utilization, scalability, cost, and system management.	08	2	2

OR

4a ✓	A company is planning to virtualize its IT infrastructure to improve resource utilization and reduce operational costs. Discuss the advantages and disadvantages of virtualization. Explain how the benefits outweigh the limitations in supporting modern cloud computing environments.	06	3	2
4b ✓	A cloud service provider intends to build an Infrastructure as a Service (IaaS) platform using Microsoft Hyper-V. Explain the Microsoft Hyper-V architecture and discuss how it supports virtual machine creation, resource allocation, device virtualization, and secure isolation between multiple virtual machines. Support your answer with a neat diagram.	10	3	2
5a ✓	A company is migrating its on-premise relational database to AWS. Analyze the AWS storage solutions and recommend suitable services for storing structured, semi-structured and unstructured data with justification.	06	4	4
5b ✓	Evaluate different storage technologies used in data-intensive computing, including distributed file systems, storage clouds and NoSQL databases. Recommend an appropriate storage solution for a cloud-based social networking platform with justification.	10	4	5

OR

6a	A global scientific collaboration requires secure access, efficient sharing and parallel processing of petabytes of experimental data. Evaluate the suitability of Data Grids and Data Clouds for this application and recommend the most appropriate solution with suitable justification.	08	4	5
6b	A multinational software company plans to build a highly scalable cloud-native application. Analyze the architectural	08	4	4

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components of Google App Engine and Microsoft Azure, highlighting their approaches to scalability, storage and runtime management.

7a A company wants to migrate its application between AWS and Azure without significant modifications while minimizing vendor lock-in. Analyze the architectural principles and quality attributes that should be considered. 08 3 4

7b A financial institution wants employees to authenticate once and securely access resources hosted on Azure, AWS, and GCP while maintaining centralized identity management. Analyze the identity management solution that should be adopted and justify your recommendation. 08 3 4

OR

A software company is evaluating different cloud service models for its business requirements. Identify the most suitable cloud service model (IaaS, PaaS, or SaaS) for each of the following scenarios and justify your choice.

1. A company wants to create and manage virtual machines, storage, and networking while retaining full control over the operating system. 1
2. A startup wants to develop and deploy web applications without managing servers, operating systems, or middleware. 2
3. An organization wants employees to use an online Customer Relationship Management (CRM) application without installing any software. 3
4. A development team wants a managed environment that provides runtime, databases, and automatic scaling for application development. 4
5. An enterprise wants to migrate its legacy application to the cloud while managing its own operating system and application software. 5
6. A university wants faculty and students to access email, document editing, and collaboration tools through a web browser. 5

8b A healthcare organization collects patient data from wearable devices, hospital management systems, and mobile applications. Analyze how business principles, data principles, and security principles together ensure a secure and scalable Multi-cloud solution. 10 3 4 8

9a Explain the architecture and major components of an AIOps system. 06 5 2

9b Compare OWASP Top 10, CIS Controls, and MITRE ATT&CK with respect to their objectives, implementation approaches, and applicability in DevSecOps. 10 5 4

OR

10a	Explain the importance of version control in Multi-cloud DevOps and discuss the different components that should be maintained under version control.	08	5	2	8
10b	A retail organization experiences unpredictable workload spikes across AWS, Azure, and GCP, resulting in frequent performance degradation. Design an AIOps-based solution to automatically detect anomalies, perform root cause analysis, and optimize resource allocation.	08	5	3	7
					15

$$49 + 16 + 16 = 81$$

$$179 + 86$$

$$140 + 160 + 9$$